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Decoding pH-Driven Phase Transition of Lipid Nanoparticles

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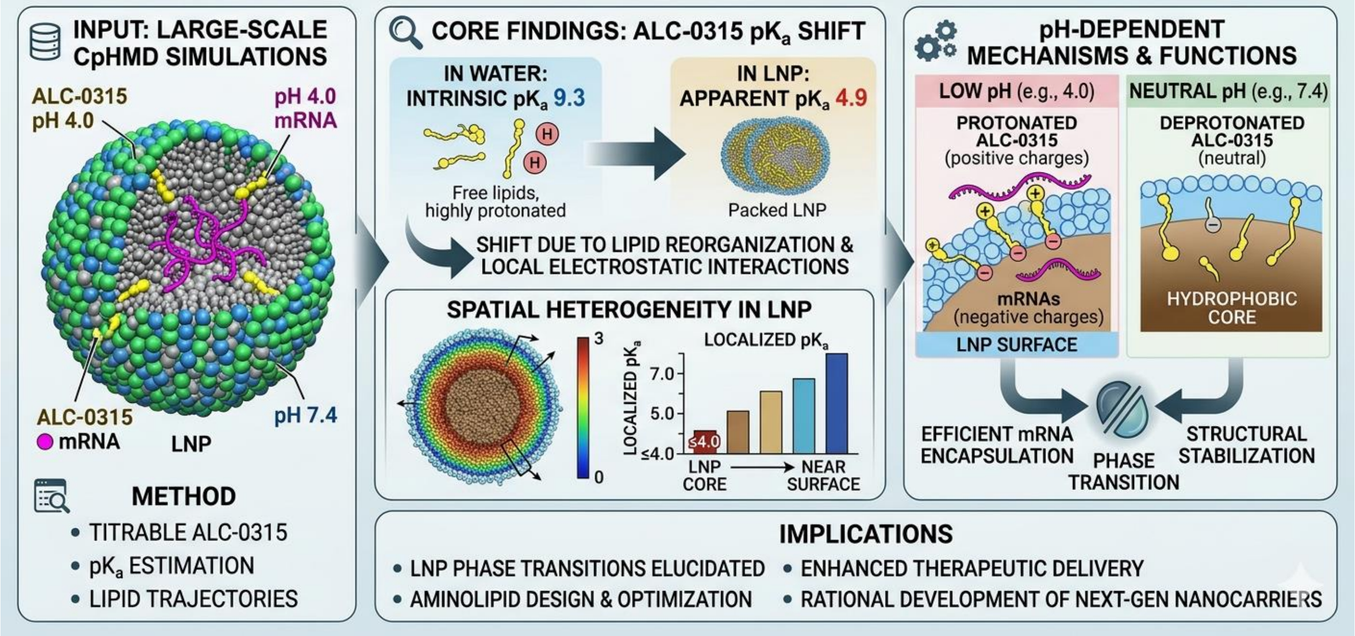
aminolipid | protonation | constant-pH molecular dynamics simulation | lipid nanoparticle | pKa

ABSTRACT

The functionality of lipid nanoparticles (LNPs) as delivery systems in mRNA-based therapeutics is intricately linked to the protonation behavior of their aminolipid components. This study employs large-scale constant-pH molecular dynamics (CpHMD) simulations to decode the environment-dependent pKa of aminolipids in the Comirnaty lipid formulation, providing a detailed view of their pH-dependent structural dynamics. Our results reveal a significant shift in the apparent pKa of the aminolipid ALC-0315, from an intrinsic value of 9.3 in water to 4.9 within the LNP environment. This shift arises from the interplay between lipid reorganization and local electrostatic interactions, resulting in distinct protonation states across the LNP core and surface. At low pH, protonated aminolipids dominate the LNP surface, promoting efficient mRNA encapsulation, whereas at neutral pH, deprotonated aminolipids migrate to the hydrophobic core, driving structural stabilization. Notably, the localized pKa of aminolipids varies significantly with their position, decreasing from near-surface regions (7 to 8) to the hydrophobic core (≤4). These findings elucidate the molecular mechanisms underpinning LNP phase transitions and highlight the key role of pKa shifts for the design of aminolipids and for optimizing LNP compositions for enhanced therapeutic delivery. This study bridges experimental observations with molecular-level insights, advancing the rational development of next-generation lipid-based nanocarriers.



The Challenge in Medicine Delivery

- **The Context**
  Lipid nanoparticles (LNPs) are microscopic fat bubbles that act as vital "delivery trucks" for mRNA medicines, such as the COVID-19 vaccines.
- **The Problem**
  To be effective, these trucks must protect the medicine in the bloodstream, but once inside a cell, they must sense a more acidic environment to break open and release their cargo.
- **The Mystery**
  Until now, scientists did not fully understand how these molecules physically rearrange themselves and change their electrical charges at the atomic level during this crucial delivery process.

The Methodology

Instead of using traditional test tubes, the researchers used advanced computer simulations to create a "virtual microscope".

They built an extremely detailed, virtual replica of the exact chemical mixture used in the Pfizer-BioNTech (**Comirnaty**) vaccine.

They exposed this virtual nanoparticle to different levels of acidity (pH) to observe how thousands of individual molecules move, react, and reorganize themselves over time.

Methodology – 1
Constant-pH Molecular Dynamics (CpHMD) and lambda-Dynamics Solution

To investigate the environment-dependent protonation behavior of lipid nanoparticles at an atomistic resolution, this study employs a highly scalable constant-pH molecular dynamics (CpHMD) approach based on lambda-dynamics. The in-silico simulations were conducted using a modified version of the **GROMACS** molecular dynamics software, which integrates a recently published constant-pH code.

Additionally, the **pHbuilder** tool was utilized to generate all necessary structural and initial topology files for the system's parameterization.

Methodology - 2

- While traditional CpHMD methods can be computationally expensive, the algorithm utilized in this study provides an efficient solution by **interpolating partial charges rather than entire potential energy functions**. It uses a continuous lambda-coordinate to transition smoothly between the fully protonated (lambda = 0) and fully deprotonated (lambda = 1) states of each titratable group.
- This specific mathematical approach drastically reduces computational overhead, successfully **allowing for the simultaneous simulation of hundreds of titratable aminolipid sites on a microsecond timescale**.

Methodology - 3

- To accurately drive these shape-shifting dynamics, three specific lambda-dependent potential energy functions were added to the total Hamiltonian (energy state) of the system: a pH-dependent potential, a biasing potential, and a macroscopic correction potential.
- The *a priori* unknown polynomial coefficients required for the correction potential were derived beforehand via thermodynamic integration. Throughout these constant-pH simulations, standard physiological parameters were maintained by controlling the temperature at 310 K using a velocity-rescale algorithm and the pressure at 1 bar using a C-rescale algorithm.

Key Results - 1

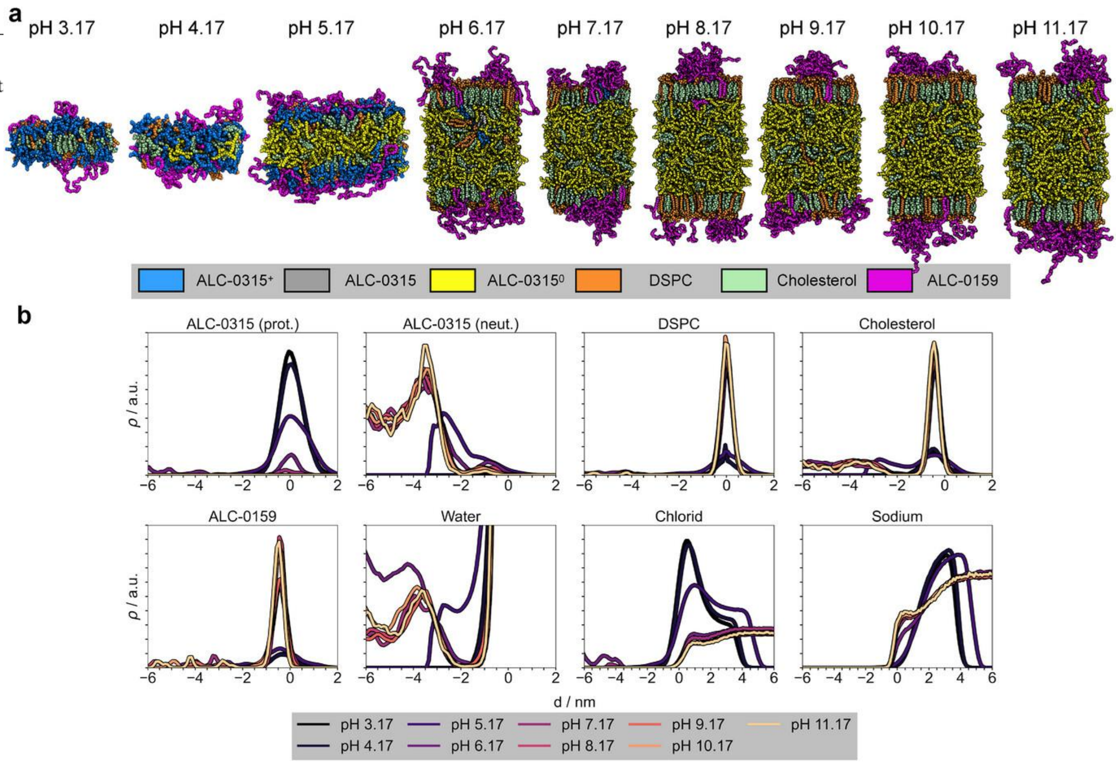
A Complete Reorganization
The study found that these nanoparticles literally reorganize their internal architecture depending on their environment.

In Neutral Environments
(the bloodstream), specific molecules shed their electrical charge and migrate to the center, creating a stable, oil-like core that securely packs the mRNA.

In Acidic Environments
(inside of a cell), the molecules at the surface become electrically charged, which triggers the nanoparticle to shift its structure and help release the medicine.

TABLE 1 | Systems studied with all-atom molecular dynamics simulations using the constant-pH method. The number of solvent molecules represents always the final topology. (†ALC-0315, ‡ALC-0315$^{\text{Re-evaluated}}$).

System	ALC-0315:DSPC:Chol: ALC-0159:mRNA	$H_2O:Na^+:Cl^-$:Buf	Sim. Time (μs)	pH	Replicas	Objective
A	1:0:0:0:0	6938:19:20:1	1.7	—	25	Thermodynamic Integration
B‡	1:0:0:0:0	6937:19:20:3	1.0	9.26	10	Goodness of polynomial coefficient
C‡	1:0:0:0:0	6937:19:20:3	1.0	4.17-14.17	11x10	Intrinsic pK_a
D	122:24:112:6:0	21660:58:180:245	1.0–5.0	5.17-11.17	7x1	Apparent pK_a
E	122:24:112:6:0	21782:58:119:123	1.0–2.4	5.17-11.17	7x1	Apparent pK_a
F^1	488:96:448:24:0	47958:130:287:90	6.7	6.17	1	Apparent pK_a (2x2)
F^2	488:96:448:24:0	45122:122:171:90	7.9	7.17	1	Apparent pK_a (2x2)
F^3	488:96:448:24:0	37315:101:137:60	8.6	8.17	1	Apparent pK_a (2x2)
F^4	488:96:448:24:0	40702:110:110:60	8.2	9.17	1	Apparent pK_a (2x2)
F^5	488:96:448:24:0	35871:97:97:60	8.6	10.17	1	Apparent pK_a (2x2)
F^6	488:96:448:24:0	34255:93:93:60	8.2	11.17	1	Apparent pK_a (2x2)
G	488:96:448:24:4	49512:120:148:60	1.91.9	7.17	2	Interaction with mRNA (2x2)
D‡	122:24:112:6:0	21660:45:167:1521660:56:167:1521660:58:167:245	7.86.85.5	3.17-5.17	3x1	Apparent pK_a
$F^{0‡}$	488:96:448:24:0	51365:139:375:9051668:140:344:90	5.24.2	5.17	2x1	Apparent pK_a (2x2)
$F^{1‡}$	488:96:448:24:0	47958:130:287:90	10.2	6.17	1	Apparent pK_a (2x2)
$F^{2‡}$	488:96:448:24:0	45122:122:171:90	5.5	7.17	1	Apparent pK_a (2x2)
$F^{3‡}$	488:96:448:24:0	37315:101:137:60	5.8	8.17	1	Apparent pK_a (2x2)
$F^{4‡}$	488:96:448:24:0	40702:110:110:60	5.1	9.17	1	Apparent pK_a (2x2)
$F^{5‡}$	488:96:448:24:0	35871:97:97:60	5.6	10.17	1	Apparent pK_a (2x2)
$F^{6‡}$	488:96:448:24:0	34255:93:93:60	5.4	11.17	1	Apparent pK_a (2x2)
G‡	488:96:448:24:4	49512:120:129:6049491:139:128:60	10.110.0	7.17	2	Interaction with mRNA (2x2)
$H^{0‡}$	122:24:112:6:0	21500:56:156:15	2.8	4.17	1	Interaction with TNS (10)
$H^{1‡}$	488:96:448:24:0	47276:130:129:90	4.4	6.17	1	Interaction with TNS (41)
$H^{2‡}$	488:96:448:24:0	44491:122:85:90	5.3	7.17	1	Interaction with TNS (41)
$H^{3‡}$	488:96:448:24:0	35214:97:56:60	1.4	11.17	1	Interaction with TNS (41)

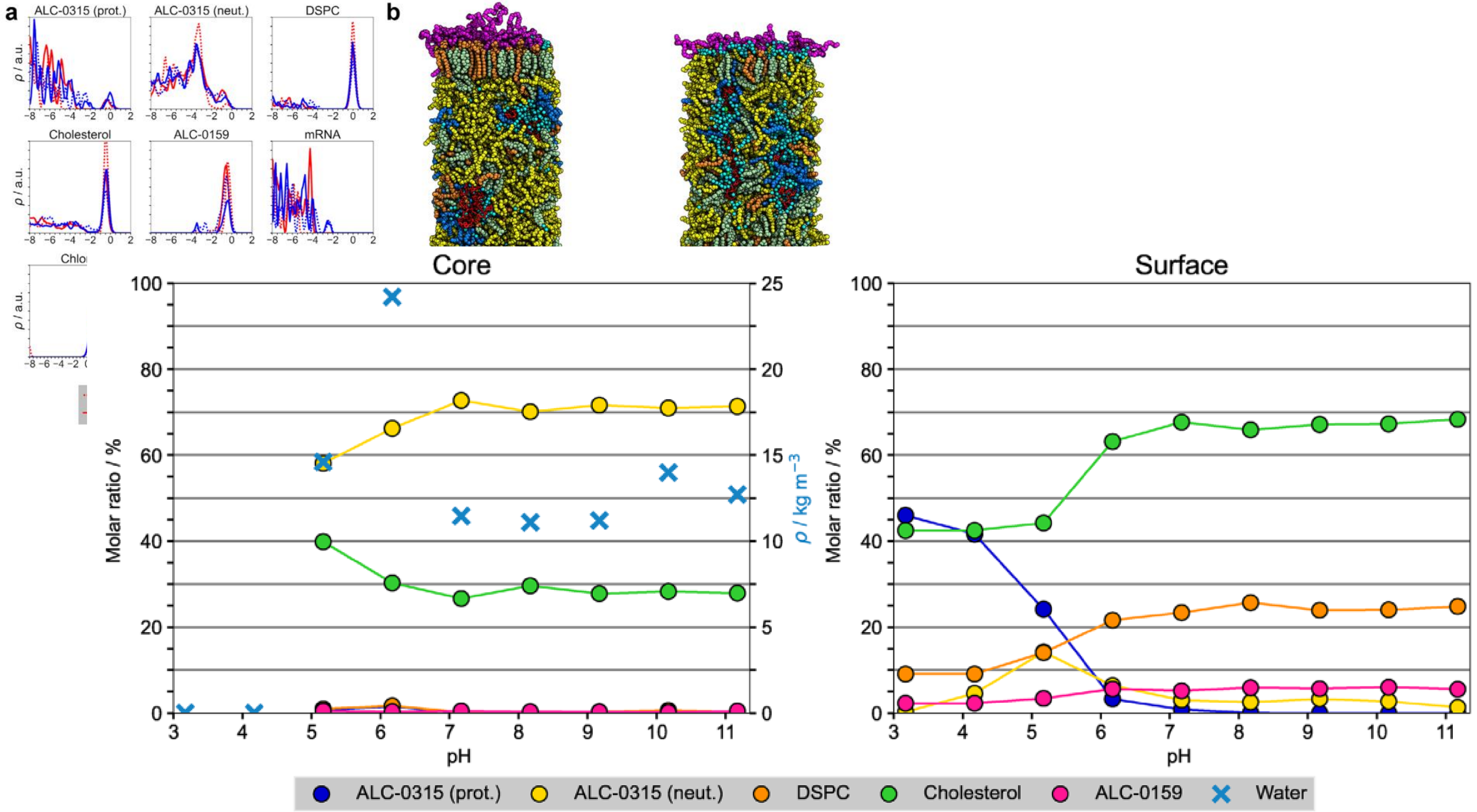


Key Results -2

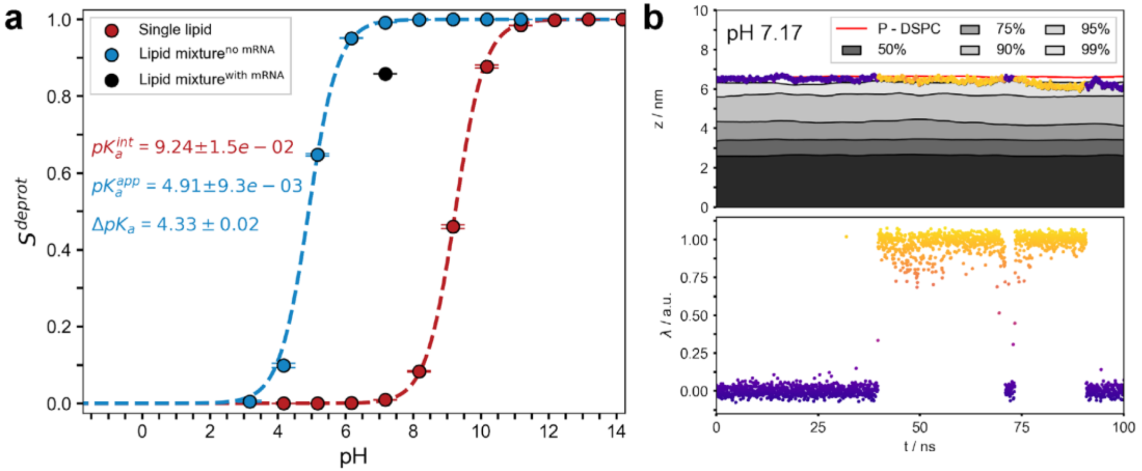
Different Rules for Different Areas: A major discovery was that the molecules in the nanoparticle do not all act the same way at the same time.

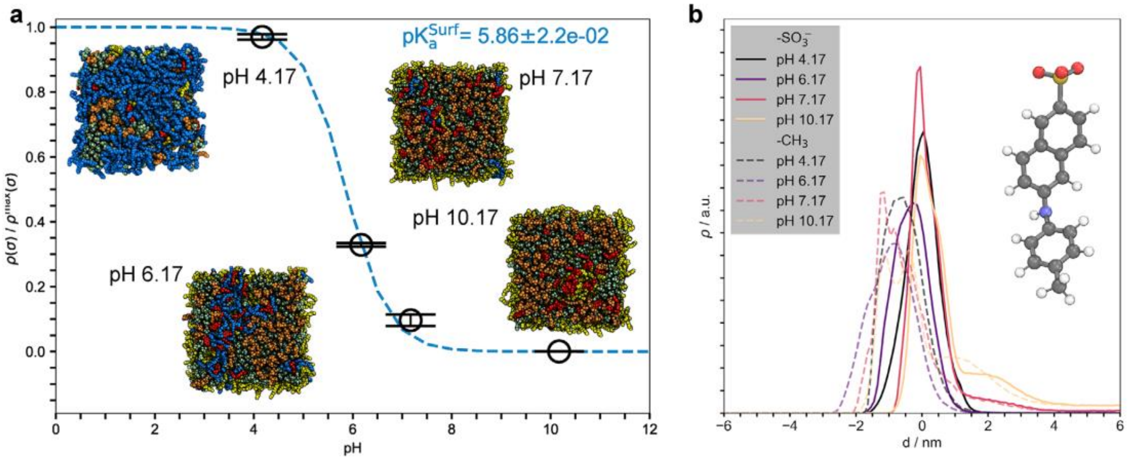
Surface vs. Core: A molecule's ability to react to its environment depends entirely on where it is located.

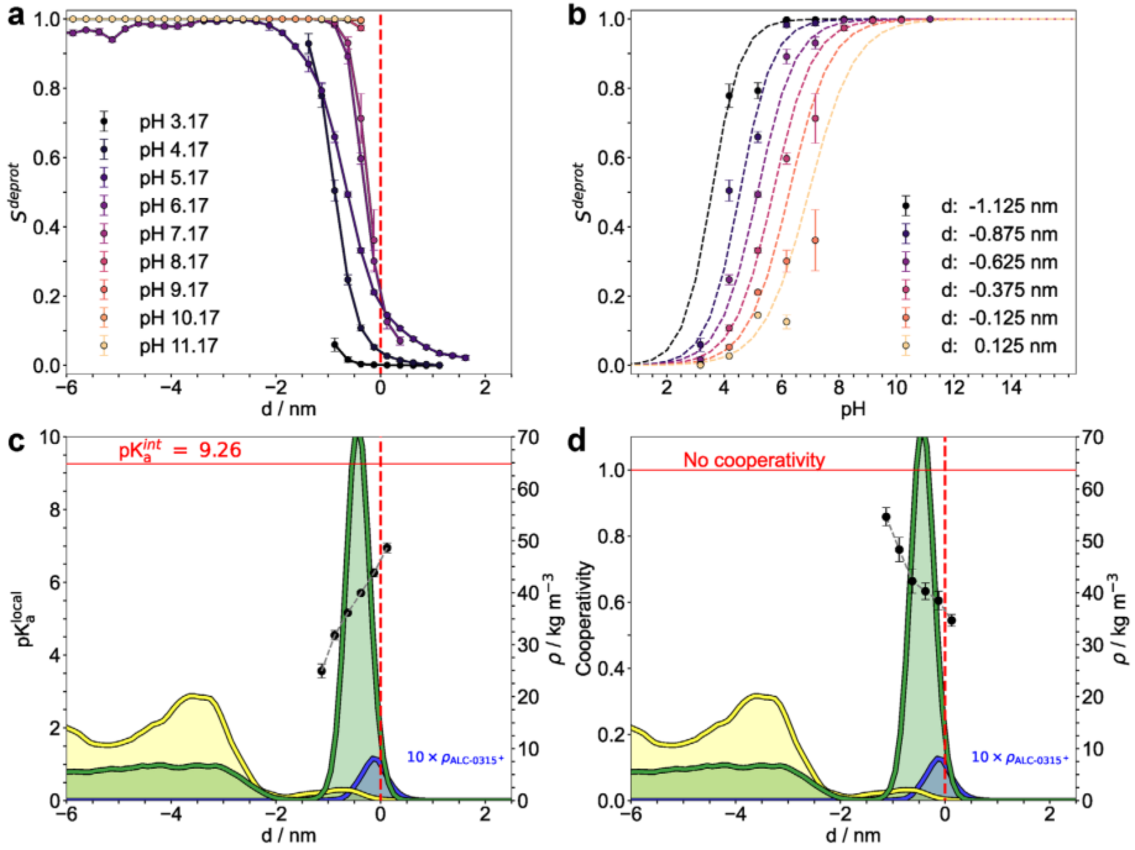
The Insight: Molecules sitting at the outer boundary of the nanoparticle gain electrical charges much more easily than the exact same molecules hidden deep inside the core.



a
ALC-0315 (prot.)
ALC-0315 (neut.)
DSPC
Cholesterol
ALC-0159
mRNA
ρ / a.u.
b
Core
Surface
Molar ratio / %
ρ / kg m⁻³
pH
ALC-0315 (prot.)
ALC-0315 (neut.)
DSPC
Cholesterol
ALC-0159
Water


a
Single lipid
Lipid mixture^no mRNA
Lipid mixture^with mRNA
pK_a^{int} = 9.24±1.5e − 02
pK_a^{app} = 4.91±9.3e − 03
ΔpK_a = 4.33 ± 0.02
S^{deprot}
pH
b
pH 7.17
P - DSPC
50%
75%
90%
95%
99%
z / nm
λ / a.u.
t / ns


a
pK_a^{Surf}= 5.86±2.2e-02
pH 4.17
pH 7.17
pH 10.17
pH 6.17
ρ(σ) / ρ^max(σ)
pH
b
$-SO_3^-$
pH 4.17
pH 6.17
pH 7.17
pH 10.17
$-CH_3$
pH 4.17
pH 6.17
pH 7.17
pH 10.17
ρ / a.u.
d / nm


a
S^{deprot}
pH 3.17
pH 4.17
pH 5.17
pH 6.17
pH 7.17
pH 8.17
pH 9.17
pH 10.17
pH 11.17
d / nm
b
S^{deprot}
d: -1.125 nm
d: -0.875 nm
d: -0.625 nm
d: -0.375 nm
d: -0.125 nm
d: 0.125 nm
pH
c
pK_a^{int} = 9.26
pK_a^{local}
ρ / kg m^{-3}
10 × $\rho_{ALC\text{-}0315^+}$
d / nm
d
No cooperativity
Cooperativity
ρ / kg m^{-3}
10 × $\rho_{ALC\text{-}0315^+}$
d / nm

Importance for **Nanobiomaterials**

- **A New Perspective**
  Previously, the field often treated these nanoparticles as static objects with a single, uniform response to acidity.
- **Connecting the Dots**
  This research proves the nanoparticles are highly dynamic and successfully bridges the gap between macroscopic laboratory experiments and microscopic atomic behavior.
- **A Better Blueprint**
  It finally explains the mechanical reasons why these specific lipid materials are so exceptionally good at delivering genetic material.

What Scientists Can Achieve Next

Less Guesswork
Researchers no longer have to rely purely on trial-and-error to test new material combinations for drug delivery.

Custom Designs
Using these insights, scientists can now rationally design custom, next-generation nanocarriers whose shape-shifting abilities are perfectly tuned for specific targets.

The Future of Medicine
This deep understanding will accelerate the creation of highly efficient, durable treatments, including new vaccines for cancer and infectious diseases, and targeted gene therapies.

Thank you for your time and attention! Are there any questions?

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